

Minhwa Lee

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AREAS OF INTEREST

Health AI agent, LLM evaluation, human-AI collaboration, and the compute-efficient deployment of trustworthy AI systems in healthcare.

EDUCATION

M.S., Computer Science, University of Massachusetts Amherst, MA Sep 2021 – May 2023

B.A., Computer Science & Mathematics, The College of Wooster, OH Aug 2017 – May 2021

TECHNICAL SKILLS

- **Languages & ML Frameworks:** Python, R, SQL, PyTorch, Hugging Face Transformers
- **AI/LLM Tools:** LangChain, LangSmith, LlamaIndex, Google Vertex AI, OpenAI API
- **Cloud & Deployment:** AWS, GCP, Azure DevOps, Docker, CI/CD, Streamlit
- **Other:** Git, Power BI, PostgreSQL, JavaScript, Healthcare Standards (FHIR, SNOMED-CT, ICD-10)

PUBLICATIONS

* denotes equal contribution. See [Google Scholar](#) for the most recent updates.

Journal

- Y. Zhang*, M. Lee*, J. B. Gibbons, H. Y. Chen, Z. Yao, Y. Wang, O. Bennett, F. Ouyang, D. Levy, K. L. Tucker, and H. Yu. “Socioeconomic, Demographic and Geographic Disparities in Accessibility to Food Pantries in the United States.” *Scientific Reports*, 2026.

Conference Proceedings

- K. C. Le*, L. Wang*, M. Lee*, R. Volkov, L. T. Chau, D. Kang. “ScholaWrite: A Dataset of End-to-end Scholarly Writing.” *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.
- S. A. Hayati, M. Lee, D. Rajagopal, and D. Kang. “How Far Can We Extract Diverse Perspectives from Large Language Models?” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- R. Koo, M. Lee, V. Raheja, J. I. Park, Z. M. Kim, and D. Kang. “Benchmarking Cognitive Biases in Large Language Models as Evaluators.” *Annual Meeting of the Association for Computational Linguistics (ACL) Findings*, 2024.
- V. Deshpande, M. Lee, Z. Yao, Z. Zhang, J. B. Gibbons, and H. Yu. “LocalTweet to LocalHealth: A Mental Health Surveillance Framework Based on Twitter Data.” *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, 2024.
- S. Kwon, R. Garodia, M. Lee, Z. Yang, and H. Yu. “Vision Meets Definitions: Unsupervised Visual Word Sense Disambiguation Incorporating Gloss Information.” *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Workshop Proceedings

- M. Lee, Z. M. Kim, V. Khetan, and D. Kang. “Human-AI Collaborative Taxonomy Construction: A Case Study in Profession-specific Writing Assistants.” *3rd Workshop of Intelligent and Interactive Writing Assistants, ACM Conference on Human Factors in Computing Systems (CHI)*, 2024.

Preprints

- **M. Lee et al.** "Deploying a Human-centered, Multi-Agent AI Clinical Trial Matching System: A Real-World Case Study at a Large U.S. Academic Medical Center." *Under Review*, 2026.
- D. Das*, K. D. Langis*, A. Martin*, J. Kim*, **M. Lee***, Z. M. Kim*, S. A. Hayati, R. Owan, B. Hu, R. S. Parkar, R. Koo, J. I. Park, A. Tyagi, L. Ferland, S. Roy, V. Liu, and D. Kang. "Under the Surface: Tracking the Artifactuality of LLM-Generated Data." *arXiv*, 2024.

PROFESSIONAL EXPERIENCE

Senior Software Engineer (AI Engineer), Mayo Clinic (via CitiusTech Inc.)

Apr 2025 – Present

- **AI-driven clinical trial matching platform** — Led the design and deployment of an end-to-end system that screens patients against trial eligibility from unstructured EHR data (clinical notes, lab reports, FHIR resources), replacing brittle keyword matching with a semantic QA engine; cut compute cost per run by 95% (\$136 → \$1.50/patient) and clinician chart-review time by 80% (3 hrs → 30 min). (**Manuscript under review by NEJM AI**)
 - Architected the multi-agent screening pipeline — search, screening, critic, and reflection agents — automating end-to-end patient screening.
 - Built a FHIR-compatible, chain-of-thought extraction pipeline parsing noisy longitudinal records into structured patient case summaries.
 - Designed a human-in-the-loop validation interface for clinicians to review and correct outputs, refining performance on ambiguous cases while building a gold-standard evaluation set.
- Developed a synthetic patient-data generator for automated model evaluation in a PHI-compliant environment, enabling robust simulations despite scarce labeled clinical data.
- Led an evaluation framework probing medical multimodal models for failure modes, informing safe AI-driven clinical decision support.

Research Assistant, University of Massachusetts Lowell

Jul 2024 – Jan 2025

- Fine-tuned RoBERTa and LLaMA on US Veterans Affairs EHRs to extract Social Determinants of Health (SDOH) entities, then applied regression analysis to link SDOH with disease-outbreak patterns among veterans.
- Investigated how implicit biases (health literacy, education, demographic disparity) degrade clinical-AI diagnostic accuracy in after-visit interactions.
- Built a real-time, LLM-powered dashboard flagging suicide-risk signals from unstructured clinical notes for mental-health practitioners.

Visiting Researcher, University of Minnesota Twin Cities

Jun 2023 – Jul 2024

- Partnered with Accenture Labs to design a human-AI collaborative knowledge-construction system for expert-level enterprise LLMs (**CHI 2024 In2Writing workshop**).
- Built and released ScholaWrite, a large-scale dataset capturing the end-to-end scholarly writing process, and fine-tuned LLaMA with LoRA to power an AI scholarly writing assistant (**ACL 2026**).
- Designed and open-sourced a benchmarking suite evaluating LLM reasoning and human-value alignment, adopted widely (300+ citations). (**ACL 2024, EMNLP 2024, Preprint**)

Data Scientist Intern, Microsoft

Jan – Feb 2023

- Built an NLP pipeline to automatically extract AI/ML topics from internal project proposals (FY21–22), replacing a manual tagging process.

- Shipped a Power BI dashboard surfacing fine-tuned-model insights on AI/ML project trends for cross-functional portfolio visibility.

Graduate Student Researcher, University of Massachusetts Amherst

Aug 2021 – May 2023

- Developed an unsupervised multimodal retrieval model combining Bayesian inference with GPT-3 for visual word-sense disambiguation (**ACL 2023**).
- Built ML models predicting neighborhood-level mental-health outcomes from a curated Twitter corpus and CDC-survey signals (**LREC-COLING 2024**).
- Constructed a web-scraped geolocation dataset of U.S. food pantries and modeled socioeconomic and demographic disparities in access (**Scientific Reports**).

ACADEMIC SERVICE

- Reviewer, Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026
- Reviewer, Conference on Language Modeling (**COLM**), 2026
- Reviewer, International Conference on Learning Representations (**ICLR**), 2025, 2026
- Reviewer, American Medical Informatics Association (**AMIA**) Clinical Informatics Conference, 2025
- Reviewer, Association for Computational Linguistics (**ACL**) **Rolling Review**, 2024, 2025, 2026

AWARDS & SCHOLARSHIPS

- Finalist, ACM Student Research Competition at Grace Hopper Celebration, 2021
- Honorable Mention, MAA Student Poster Presentation, Joint Mathematical Meetings, 2021
- Magna Cum Laude, The College of Wooster, 2021
- Departmental Honors (Computer Science, Mathematics), The College of Wooster, 2021
- The Vivien Chan Prize in Interdisciplinary Sciences, The College of Wooster, 2021
- Pi Mu Epsilon Induction, The College of Wooster, 2019
- Dean's List, The College of Wooster, 2018–2021
- Grace Hopper Celebration Student Scholarship, 2021
- Wooster International Merit Scholarship (\$38,000/year), 2017–2021